Part 3. Understand the Paper Before Downloading Data

Information from the Paper	Group Answer
Title of the paper	Transcriptomic analysis reveals candidate genes associated with salinity stress tolerance during the early vegetative stage in fababean genotype, Hassawi-2
Authors	Muhammad Afzal, Salem S. Alghamdi, Muhammad Altaf Khan, Sulieman A. Al-Faifi, & Muhammad Habib ur Rahman
Year	2023
Journal	*Scientific Reports*, 13, 21223
Organism	Faba bean (*Vicia faba* L.), salt-tolerant genotype Hassawi-2
Tissue / organ / cell type	Leaves at the early vegetative/three-leaf stage
Biological question	To identify genes and molecular mechanisms involved in salinity stress tolerance in Hassawi-2 faba bean at different durations of salt exposure.
Control condition	Faba bean plants grown without the 200 mM NaCl salt treatment
Treatment or stress condition	200 mM NaCl salt stress for 6, 12, 24, 48, and 72 hours
Number of biological replicates	3 biological replicates per condition/time point
Sequencing platform	Illumina NovaSeq 6000 (sequencing outsourced to Macrogen)
Single-end or paired-end sequencing	Paired-end

Read length, if reported	101 bp per read
RNA-seq repository	NCBI Sequence Read Archive (SRA)

BioProject / Study accession	PRJNA943415 (SRA submission ID SUB12947558)
Run accession numbers	Not listed individually in the paper — retrieve the SRR list from the BioProject/SRA page for PRJNA943415
Reference genome or transcriptome used by the authors	None, de novo assembly (no reference genome used; faba bean genome had only just been published)
Main RNA-seq software or tools reported by the authors	Trimmomatic (adapter/quality trimming) → Trinity (de novo assembly) → CD-HIT-EST (clustering into unigenes) → TransDecoder (ORF prediction) → Bowtie (read alignment) → RSEM (expression/FPKM quantification) → edgeR (DEG analysis, TMM normalization) → BLASTX/DIAMOND against NR, Swiss-Prot, KEGG, GO, COG/KOG for annotation